

```
#practical 3
#TANISHQ SIJARIA
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!pip install tensorflow scikit-learn pandas matplotlib seaborn
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.20.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
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Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
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```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report
```

```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

```
data = load_breast_cancer()
```

```
X = data.data
y = data.target
```

```
print(X.shape)
print(y.shape)
```

```
(569, 30)
(569,)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = Sequential()

model.add(Dense(64, activation='relu', input_shape=(30,)))

model.add(Dense(32, activation='relu'))

model.add(Dense(1, activation='sigmoid'))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape` argument to a layer that does not require it.
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

```
history = model.fit(
    X_train,
    y_train,
    epochs=30,
    batch_size=16,
    validation_split=0.2,
    verbose=1
)
```

```
Epoch 1/30
23/23 ————— 0s 10ms/step - accuracy: 0.9643 - loss: 0.1945 - val_accuracy: 0.9780 - val_loss: 0.0899
Epoch 3/30
23/23 ————— 0s 15ms/step - accuracy: 0.9670 - loss: 0.1192 - val_accuracy: 0.9780 - val_loss: 0.0899
Epoch 4/30
23/23 ————— 0s 5ms/step - accuracy: 0.9725 - loss: 0.0899 - val_accuracy: 0.9780 - val_loss: 0.0899
Epoch 5/30
23/23 ————— 0s 5ms/step - accuracy: 0.9835 - loss: 0.0740 - val_accuracy: 0.9670 - val_loss: 0.0899
Epoch 6/30
23/23 ————— 0s 5ms/step - accuracy: 0.9863 - loss: 0.0634 - val_accuracy: 0.9670 - val_loss: 0.0899
Epoch 7/30
23/23 ————— 0s 5ms/step - accuracy: 0.9863 - loss: 0.0555 - val_accuracy: 0.9670 - val_loss: 0.0899
Epoch 8/30
23/23 ————— 0s 5ms/step - accuracy: 0.9863 - loss: 0.0498 - val_accuracy: 0.9560 - val_loss: 0.0899
Epoch 9/30
23/23 ————— 0s 6ms/step - accuracy: 0.9863 - loss: 0.0450 - val_accuracy: 0.9780 - val_loss: 0.0899
Epoch 10/30
23/23 ————— 0s 5ms/step - accuracy: 0.9890 - loss: 0.0410 - val_accuracy: 0.9670 - val_loss: 0.0899
Epoch 11/30
```

```

Epoch 13/30
23/23  0s 5ms/step - accuracy: 0.9918 - loss: 0.0313 - val_accuracy: 0.9670 - val_loss: 0.0313
Epoch 14/30
23/23  0s 5ms/step - accuracy: 0.9945 - loss: 0.0285 - val_accuracy: 0.9670 - val_loss: 0.0285
Epoch 15/30
23/23  0s 5ms/step - accuracy: 0.9918 - loss: 0.0258 - val_accuracy: 0.9670 - val_loss: 0.0258
Epoch 16/30
23/23  0s 6ms/step - accuracy: 0.9945 - loss: 0.0237 - val_accuracy: 0.9670 - val_loss: 0.0237
Epoch 17/30
23/23  0s 5ms/step - accuracy: 0.9945 - loss: 0.0215 - val_accuracy: 0.9670 - val_loss: 0.0215
Epoch 18/30
23/23  0s 5ms/step - accuracy: 0.9945 - loss: 0.0198 - val_accuracy: 0.9560 - val_loss: 0.0198
Epoch 19/30
23/23  0s 6ms/step - accuracy: 0.9945 - loss: 0.0183 - val_accuracy: 0.9451 - val_loss: 0.0183
Epoch 20/30
23/23  0s 9ms/step - accuracy: 0.9945 - loss: 0.0163 - val_accuracy: 0.9451 - val_loss: 0.0163
Epoch 21/30
23/23  0s 8ms/step - accuracy: 0.9945 - loss: 0.0149 - val_accuracy: 0.9451 - val_loss: 0.0149
Epoch 22/30
23/23  0s 10ms/step - accuracy: 0.9945 - loss: 0.0137 - val_accuracy: 0.9451 - val_loss: 0.0137
Epoch 23/30
23/23  0s 8ms/step - accuracy: 0.9945 - loss: 0.0123 - val_accuracy: 0.9451 - val_loss: 0.0123
Epoch 24/30
23/23  0s 8ms/step - accuracy: 0.9973 - loss: 0.0108 - val_accuracy: 0.9451 - val_loss: 0.0108
Epoch 25/30
23/23  0s 9ms/step - accuracy: 1.0000 - loss: 0.0100 - val_accuracy: 0.9451 - val_loss: 0.0100
Epoch 26/30
23/23  0s 13ms/step - accuracy: 0.9945 - loss: 0.0094 - val_accuracy: 0.9451 - val_loss: 0.0094
Epoch 27/30
23/23  0s 9ms/step - accuracy: 1.0000 - loss: 0.0084 - val_accuracy: 0.9451 - val_loss: 0.0084
Epoch 28/30
23/23  0s 10ms/step - accuracy: 1.0000 - loss: 0.0075 - val_accuracy: 0.9451 - val_loss: 0.0075
Epoch 29/30
23/23  0s 8ms/step - accuracy: 1.0000 - loss: 0.0068 - val_accuracy: 0.9451 - val_loss: 0.0068
Epoch 30/30
23/23  0s 10ms/step - accuracy: 1.0000 - loss: 0.0061 - val_accuracy: 0.9451 - val_loss: 0.0061

```